

BALEA LAC, Romania

WMO: 152790

Lat: 45.6039N

Lon: 24.6147E

Elev: 6795

StdP: 11.43

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	1.7	-20.0	2.4	16.9	-13.5	3.4	15.2	33.9	24.0	29.5	24.5	5.8	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	62.7	53.4	60.3	52.8	58.3	51.8	56.3	60.1	54.7	58.2	53.3	56.9	6.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.8	82.4	57.9	53.2	77.8	56.5	51.7	73.6	55.2	27.1	60.2	25.9	58.3	24.9	57.0	63.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.8	24.5	20.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.6	19.0	21.8	24.4	32.1	39.5	47.0	49.0	51.1	43.2	37.1	28.4	22.0		
	DBStd	13.50	8.99	8.24	8.51	8.33	5.81	6.41	6.07	5.91	7.77	8.59	9.15	8.87		
	HDD50	5834	962	789	793	536	329	129	94	61	223	403	648	867		
	HDD65	11086	1427	1209	1258	986	791	539	497	432	653	864	1098	1332		
	CDD50	221	0	0	0	0	2	40	62	93	20	4	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	6.6	7.0	8.4	6.9	7.5	5.5	5.8	6.1	5.1	6.3	5.8	7.4	7.3		
	PrecAvg	38.7	2.2	2.0	2.4	3.0	4.2	4.6	5.2	3.9	3.3	2.8	2.4	2.2		
	PrecMax	63.2	4.6	4.9	4.3	7.7	10.2	8.8	9.2	8.2	7.0	5.8	6.5	4.6		
(15) Precipitation	PrecMin	23.7	0.6	0.3	0.7	0.8	1.8	0.6	0.7	0.7	0.4	0.0	0.1	0.3		
	PrecStd	10.4	1.2	1.1	1.0	1.6	2.2	2.1	2.4	2.3	1.7	1.6	1.6	1.1		
	DB	38.6	42.6	46.5	51.8	56.1	64.2	65.9	65.8	64.5	57.0	48.6	48.7			
	MCWB	30.0	34.2	37.1	41.6	48.3	51.2	51.9	55.7	51.0	41.8	39.8	32.8			
	2% DB	35.1	38.2	42.0	48.4	52.9	60.6	62.0	62.3	58.1	53.5	44.7	40.5			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	28.1	32.7	35.9	40.5	45.7	52.5	53.6	54.2	50.5	40.2	35.5	30.9			
	DB	32.4	34.4	39.0	45.8	50.4	58.3	59.3	60.2	55.3	50.0	42.0	36.6			
	MCWB	28.2	30.0	33.6	38.6	44.3	51.3	53.1	53.1	49.0	39.7	34.9	31.0			
	10% DB	30.4	31.9	35.5	43.4	47.7	55.4	57.1	58.3	52.5	47.6	39.4	33.1			
	MCWB	27.4	27.9	31.9	37.5	43.8	50.4	51.8	52.1	46.5	39.9	34.8	29.0			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	32.4	36.8	40.3	44.4	50.3	57.4	57.9	58.2	54.4	47.5	42.3	37.8			
	MCDB	35.5	39.9	44.3	49.0	53.6	59.9	61.5	62.3	59.9	51.2	46.8	39.9			
	2% WB	31.0	33.5	36.8	42.1	47.4	54.2	55.8	56.7	52.3	44.7	39.6	34.9			
	MCDB	32.9	38.3	40.4	46.5	50.3	57.7	59.4	60.6	57.6	49.1	42.3	39.0			
	5% WB	29.8	31.0	34.6	40.1	45.6	52.7	54.3	55.0	49.6	43.1	38.0	32.0			
(33) Mean Daily Temperature Range	MCDB	31.9	33.9	38.1	45.1	48.7	56.4	57.5	58.5	54.1	47.8	40.1	35.0			
	10% WB	28.0	29.5	32.2	38.3	44.0	51.2	52.9	53.5	47.9	41.3	36.2	30.0			
	MCDB	29.8	31.7	35.3	42.9	47.1	55.0	55.9	57.5	51.8	45.8	38.2	32.6			
	5% DB	10.3	12.0	13.9	11.4	11.6	11.7	10.6	8.8	8.2	11.5	10.1	11.7			
	MCWBR	7.2	7.7	9.2	7.6	10.1	9.4	9.5	8.2	8.3	9.3	6.9	6.8			
(40) Clear-Sky Solar Irradiance	5% WB	8.9	10.2	11.6	10.5	10.8	10.1	8.9	7.8	7.8	10.2	7.0	10.3			
	MCWBR	6.8	7.3	9.4	7.5	9.7	9.3	9.0	8.6	7.8	9.6	6.3	8.7			
	taub	0.234	0.253	0.280	0.318	0.310	0.312	0.322	0.328	0.300	0.276	0.259	0.236			
	taud	2.455	2.436	2.461	2.422	2.471	2.481	2.444	2.421	2.498	2.572	2.571	2.500			
	Ebn at Noon	288	301	305	299	304	302	298	292	293	289	277	276			
(44) All-Sky Solar Radiation	Edh at Noon	21	26	30	34	34	34	35	34	29	23	19	18			
	RadAvg	441	702	1040	1355	1651	1787	1813	1645	1203	834	488	357			
	RadStd	46	87	110	149	163	163	158	141	120	106	50	43			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)		+1.12	N/A	N/A	+1.40	N/A	N/A	-223	-315	+144	+45

Nomenclature: See separate page